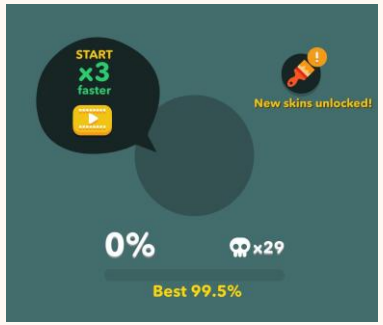


KNN FOR MOVIE RECOMMENDATIONS

HOW EFFECTIVELY CAN A KNN ALGORITHM GENERATE PERSONAL FILM RECOMMENDATIONS?

Group 14

DATA SET

GROUPLENS RESEARCH PROJECT FROM THE UNIVERSITY OF MINNESOTA

- 100k entries

PANDAS' PIVOT TABLE TO RESTRUCTURE THE DATASET

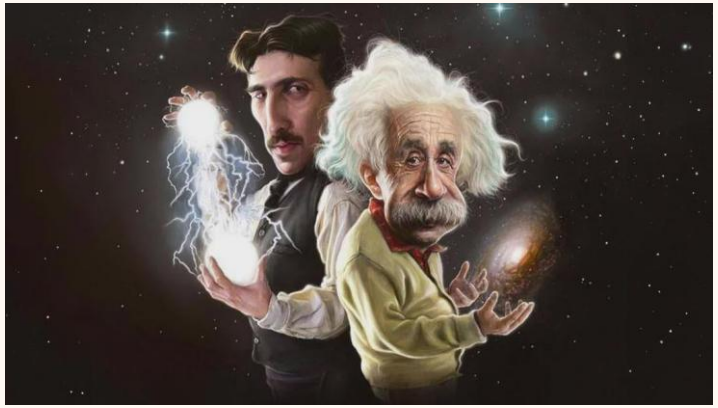
- Row represents a user id
- Column represents a movie id
- Cell values hold ratings

User ID	Movie ID	Rating
196	242	3
186	302	3
22	377	1
186	51	2

User ID	242	302	377	51
196	3	0	0	0
186	0	3	0	2
22	0	0	1	0

KNN ALGORITHM

FINDING MOVIES BASED ON USERS WITH SIMILAR TASTE TO YOU



1. **SETUP MATRIX**
   - 943 Users (rows)
   - 1682 Movies (columns)

2. **REPRESENT ACTIVE USER AND MATRIX USERS AS A VECTORS**

3. **MEASURE SIMILARITY AMONG USERS**
   - Compute similarity between active user ratings and all other users
   - Higher score = **similar taste**

$$Cos(u, v) = (u \cdot v)/(||u|| \cdot ||v||)$$

   - 1 = Exactly similar taste, same direction vectors
   - 0 = totally unrelated, perpendicular vectors
   - -1 = opposite taste, opposite vectors

4. **SORT SIMILARITY SCORE AND KEEP TOP K NEIGHBORS**

5. **RETURN NEAREST EXISTING USERS TO USER RESPONSE**

PIPELINE

1
QUIZ

Active user rates 10 random movies they have seen 1-5

2
SUBSET

Full movie data frame is split

only rated movies and corresponding user ratings

3
KNN

KNN find user ID with closest matching taste

4
RECOMMEND

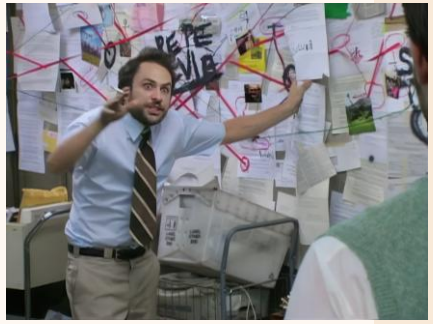
Top movies rated by best match user and not rated by active user are found

5
RESULT

Return top movie and ask user for feedback on recommendation

```
Question 1 (0 if you havent seen it)
Rating of Sabrina (1995):  4
Question 2 (0 if you havent seen it)
Rating of Restoration (1995):  3
Question 3 (0 if you havent seen it)
Rating of Conan the Barbarian (1981):  1
Question 4 (0 if you havent seen it)
Rating of Evita (1996):  0
Question 5 (0 if you havent seen it)
Rating of Four Weddings and a Funeral (1994):  5
Question 6 (0 if you havent seen it)
Rating of Die Hard (1988):  5
Question 7 (0 if you havent seen it)
Rating of Nightmare on Elm Street, A (1984):  4
Question 8 (0 if you havent seen it)
Rating of Mighty Aphrodite (1995):  1
Question 9 (0 if you havent seen it)
Rating of Willy Wonka and the Chocolate Factory (1971):  3
Question 10 (0 if you havent seen it)
Rating of Blues Brothers, The (1980):  0
```

Results from quiz



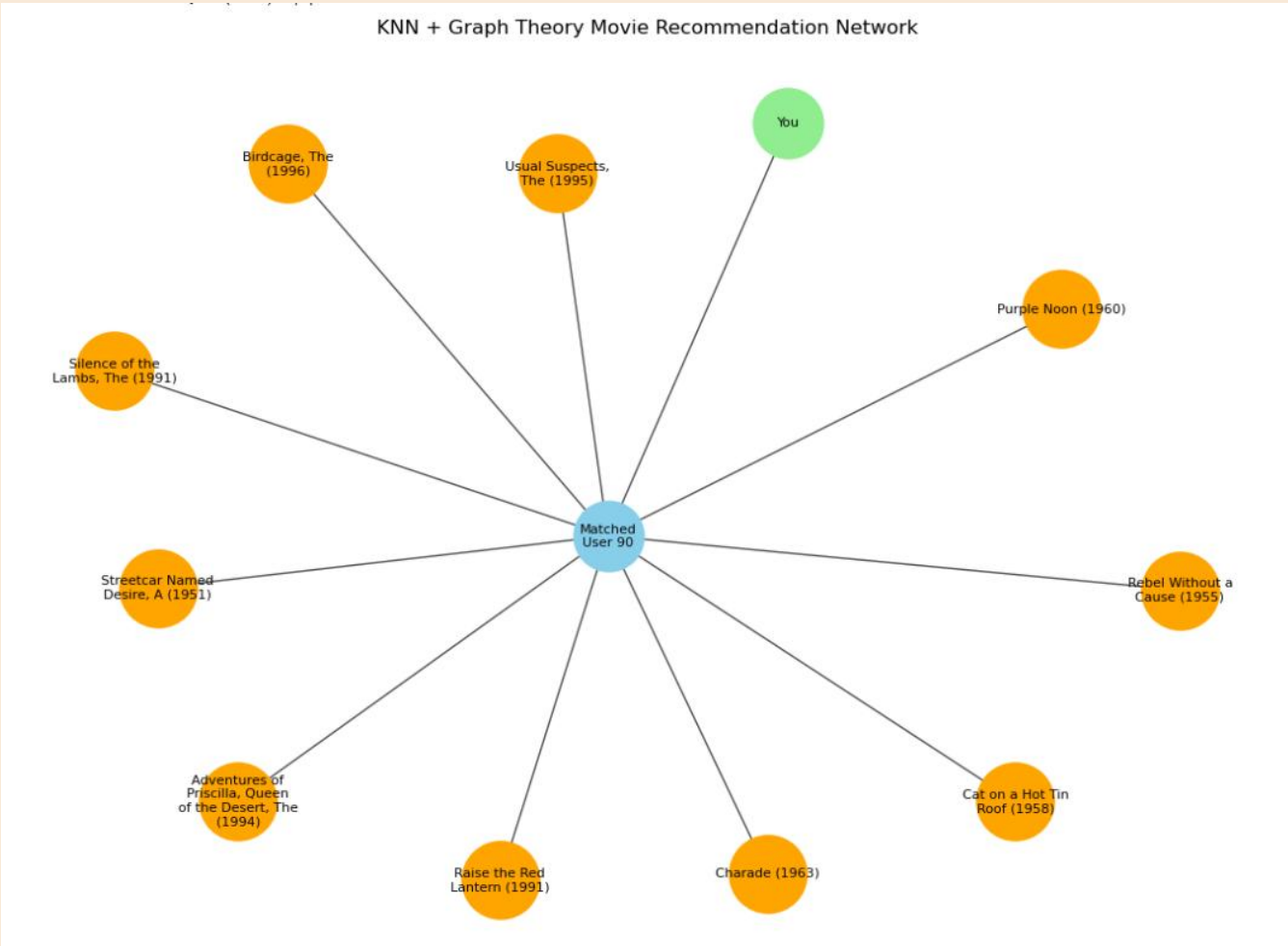
GRAPH THEORY

```
Closest User Found by KNN: 90
Similarity Score: 0.931

Graph-Based Recommendations:

Adventures of Priscilla, Queen of the Desert, The (1994)  | predicted interest: 5
Silence of the Lambs, The (1991)  | predicted interest: 5
Purple Noon (1960)  | predicted interest: 5
Birdcage, The (1996)  | predicted interest: 5
Rebel Without a Cause (1955)  | predicted interest: 5
Raise the Red Lantern (1991)  | predicted interest: 5
Usual Suspects, The (1995)  | predicted interest: 5
Charade (1963)  | predicted interest: 5
Cat on a Hot Tin Roof (1958)  | predicted interest: 5
Streetcar Named Desire, A (1951)  | predicted interest: 5
```

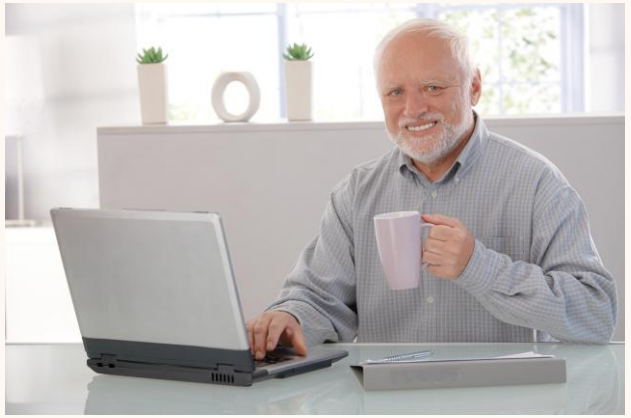
Results from Graph Theory

Results from KNN

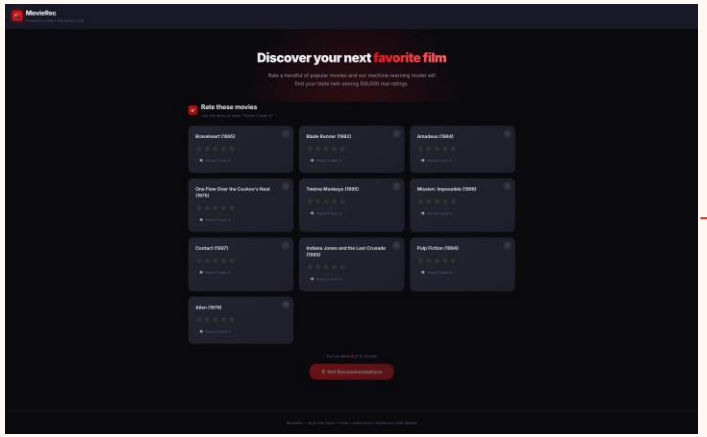
```
your best match is user:90
```

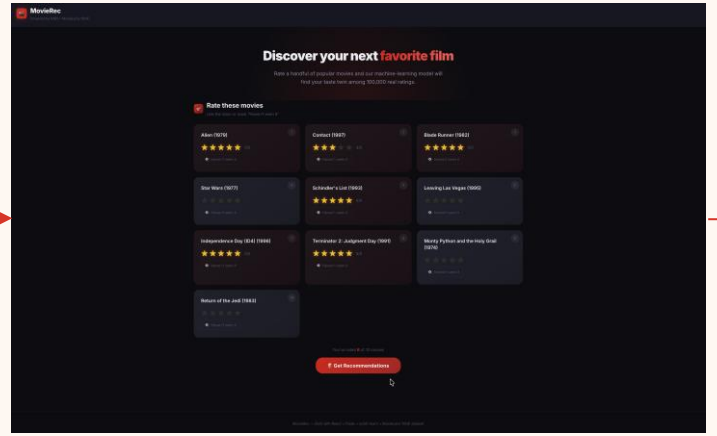


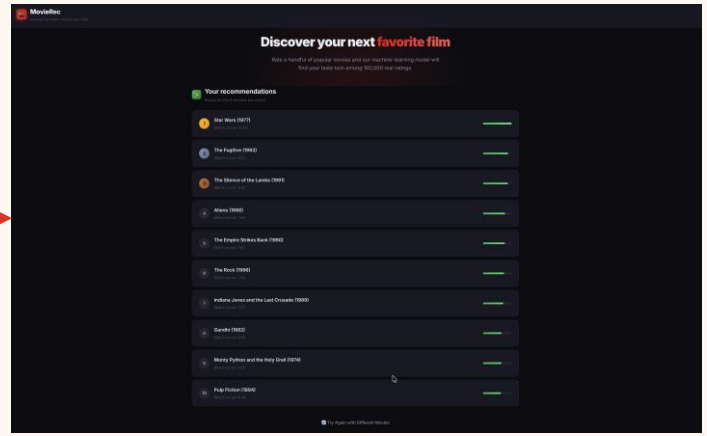
FRONTEND



- Single-paged application using React, Flask, and Vite
- Displays movies for user to rank between 0-5; if 0 is chosen, frontend replaces with a new movie from Flask movie database
- User can rate up to 10 movies for the recommendation algorithm to function
- Includes loading animations while KNN and frontend-backend data parsing is occurring
- Closest movies to your ratings & closeness score from graph theory displayed in new screen with all results







DEMO

RESULTS

- TESTING EFFICACY OF ALGORITHM -> SHIFT AROUND PARAMETERS TO GET HIGHEST SUCCESS
- MADE A SIMULATION IMITATING USER TAKING QUIZ RATHER THAN REAL WORLD DATA
- TAKE RANDOM USER, SORT BY HIGHEST RATED MOVIES (>= 4), CREATE VECTOR WITH 10 MOVIES
- RESULT IS LIST OF RECOMMENDED MOVIES
  - IF MOVIES FROM LIST MATCHES A MOVIE ORIGINAL USER HAS WATCHED – SUCCESS
  - IF NOT - FAILURE

- 20 trials per user
- **10** questions on quiz
- 4 star ratings or higher
- 20 recommendations outputted

Overall:

- Eligible users: 3283
- Executed trials: 65660
- Successes: 38013
- Failures: 27647
- Success rate: 57.89%

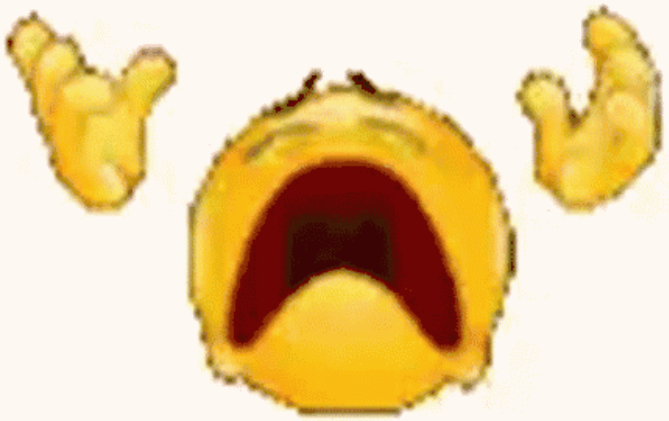
- 20 trials per user
- **20** questions on quiz
- 4 star ratings or higher
- 20 recommendations outputted

Overall:

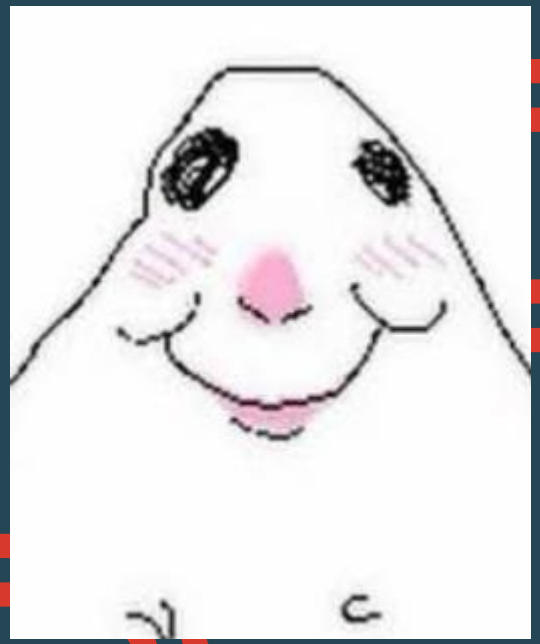
- Eligible users: 2424
- Executed trials: 48480
- Successes: 30922
- Failures: 17558
- Success rate: 68.78%

More successful! 😊

ISSUES & FUTURE WORK



- ACCOMMODATING FOR RATING LISTS IN VARIOUS SIZES
- IMPLEMENTING MODEL INTO FRONTEND TO DEPLOY ON VERCEL
- DISPLAY MOVIE POSTERS NEXT TO MOVIE NAMES ON WEBSITE
- FILTERING RESULTS FOR SPECIFIC PARAMETERS (ACTOR/ACTRESS, DIRECTOR, YEAR MADE, ETC.)
- WEIGHTING MOVIES BY POPULARITY (BETTER SELECTION POOL AND RECOMMENDING LESSER KNOWN "HIDDEN GEMS")



KNN is quite decent for recommending films. 👍

Further refined by increasing quiz & dataset size. 📈

CONCLUSIONS